UrbanStyle Project Plan

Due: in class we discuss

1) Kaggle Dataset

Recommended dataset matching the required fields (Transaction ID, Customer ID, Gender, Age, Product Category, Price, Quantity, Total Amount):

https://www.kaggle.com/datasets/mohammadtalib786/retail-sales-dataset

Interesting Simulation project for your review:

https://github.com/KG0116/store-sim

2) Step-by-step Project Tasks (UrbanStyle Business Case)

UrbanStyle = a fashion retailer. All tasks focus on this business. Pygame is obligatory for interactivity (buttons/sliders), but the central focus is on algorithms (search, forecasting, RL, GAN).

Task A — Customer Pathfinding & Store Layout (Search Algorithms)

Goal: Show how customers move inside the store and identify crowded spots.

Required:

1. Represent the store as a grid (e.g., 8×6 cells).
2. Map product categories to sections.
3. Customers enter and move to product sections.
4. Implement at least two search algorithms (BFS and A*).
5. Visualize high-traffic areas.

Pygame component: Display the grid, moving customers, and allow sliders/buttons to change algorithm or number of customers.

Optional: Add heatmap overlay or test different layouts.

Task B — Demand Forecasting (Time-Series with Small LSTM)

Goal: Predict future sales to support inventory decisions.

Required:

1. Select 5–8 categories.
2. Preprocess as weekly time-series.
3. Train a small LSTM or encoder-decoder.

4. Predict next week's sales.
5. Evaluate with MAE and RMSE.

Pygame component: Show forecast plot, with sliders to select category or training length.

Optional: Add moving average baseline or CSV export of predictions.

Task C — Dynamic Pricing with Reinforcement Learning (RL + Gym)

Goal: Adjust prices dynamically to maximize revenue.

Required:

1. Define a Gym environment (state: price, segment, recent sales; actions: adjust price; reward: revenue).
2. Use stable-baselines3 (DQN or PPO) to train an agent.
3. Run episodes and measure revenue.
4. Compare to static pricing.

Pygame component: Display live revenue plot, allow sliders to set initial price or reset environment.

Optional: Add multiple products or test multiple RL algorithms.

Task D — Simple GAN for Promotion Images

Goal: Create new promotional fashion images.

You can use MNIST-Fashion (https://github.com/zalandoresearch/fashion-mnist)

Required:

1. Train a very small GAN (e.g., DCGAN) on product images.
2. Generator: noise → image (28×28).
3. Discriminator: classify real/fake.
4. Train a few epochs and save outputs.

Pygame component: Button to generate images, sliders for noise seed.

Optional: Conditional GAN (cGAN) or combine images into promo poster.

3) Evaluation Rubric

Task	Weight	Required criteria	Optional (bonus)
Task A — Pathfinding	15%	Grid model, BFS & A* implemented, paths visualized, high-traffic zones	Heatmap overlay, layout experiments

		identified	
Task B — Forecasting	20%	Time-series prep, LSTM model runs, MAE & RMSE reported	Baseline comparison, CSV export
Task C — RL Pricing	25%	Gym env built, RL agent trained (stable-baselines3), revenue vs static shown	Multi-product or multiple RL algorithms
Task D — GANs	20%	Small GAN trained, sample outputs generated	Conditional GAN or promo poster
Pygame integration	10%	Buttons/sliders for interactivity, at least one visualization per task	Extra UI polish
Report & Code	10%	Clear report, reproducible code, short presentation	—